Ep. 381: Tomasz Nadrowski on Energy Chokepoints, Asymmetric Warfare, and Critical Minerals

Podcast: Macro hive

AI Executive Summary

1. Core Thesis The episode argues that the global order is undergoing a structural transformation from US-led globalization to a multipolar landscape characterized by asymmetric warfare and resource security challenges. Stability is no longer maintained through traditional hegemony but is increasingly fragile due to cheap, high-tech disruption capabilities (e.g., ROVs/AUVs) used by revisionist powers like China and Russia. Consequently, Western nations are pivoting toward reindustrialization in critical minerals and energy refining despite high interest rates, driven more by strategic imperatives than capital availability. Philosophically, the discussion extends to Artificial Intelligence, which is viewed not as a purely hostile or benevolent force, but as an "interlocutor" that may inadvertently reconstruct shared human assumptions (the background *Lebenswelt*) by bypassing traditional social fragmentation.

2. Key Arguments & Data

3. **Asymmetric Warfare vs. Traditional Naval Power:** The maintenance of global status quo (freedoms of navigation/trade) is becoming prohibitively costly for the West due to US relative power loss and technological advancement. Revisionist powers utilize cheap, deniable asymmetric tools like Remotely Operated Vehicles (ROVs), Autonomous Underwater Vehicles (AUVs), and anti-submarine mines to disrupt critical flows (pipelines, cables) without direct state involvement. This theory parallels the German U-boat strategy of WWI or the Russian Navy’s Murmansk-based Atlantic disruption, moving beyond Alfred Thayer Mahan's "command of the sea" to Julian Corbett's "influence on main lines."
4. **Energy Market Dynamics:** The market has shifted focus from crude oil reserves (which were not enough to cause a global crash due to China's Strategic Petroleum Reserves) to refined products. A critical bottleneck is the lack of Western refining capacity resulting from years prioritizing green energy sources over capital expenditure on infrastructure.

    - **Refining Geography:** Capacity deficits exist in Europe, but Russia and China are unreliable players (China restricts exports during tensions; Russia faces its own security constraints). The future lies in the Western Hemisphere (Brazil, US Gulf Coast) for supply, though this does not easily solve Northeast Asian access issues regarding the Panama Canal.
5. **Critical Minerals & Geopolitics:** China holds a "valve-like" leverage over critical minerals and rare earths.
    - **The "Option" Theory:** China treats mineral supply restrictions as a long-term option, increasing volatility to force concessions from the US. The US recently rejected a proposed two-year extension on critical mineral restrictions by the Chinese government (which aligned with Trump's potential term) due to DFARS rule tightening, shortening the leverage window for China.
    - **Game Theory Shift:** China's use of supply leverage shifted the conflict dynamic from a "Game of Chicken" to a "Bully game." The US had no choice but to yield initially because downstream pain thresholds in the US industrial base were higher than China's. However, on the civilian side, the US remains subject to this bully game via DFARS restrictions.
6. **Ukraine-Russia Gray Zone Tactics:** Russia employs "implausible deniability" tactics (GPS jamming, cable cutting) to normalize aggression and test Western alliance thresholds without immediate kinetic response.
    - **Strategic Objective:** Russia seeks to restore a pre-1997 sphere of influence (including Finland and Poland), viewing Ukraine as part of a broader containment strategy against NATO. This tactic exploits the "paradox of weak power," where traditional fear of Soviet strength has faded, making it harder for Western publics to prioritize security over economic concerns.
7. **Economic Reality:** Interest rates are not the primary driver of capital expenditure (CapEx) cycles; security needs are driving the rebuild of Western industrial capacity. Europe and other regions are investing heavily in defense and energy infrastructure despite fiscal strains, partly funded by public money.

8. **AI & Society (Habermasian Lens):** The speaker views AI through the lens of communicative action (*Lebenswelt* vs. System).
    - **AI as Interlocutor:** Rather than just a tool colonizing life, AI functions as an interlocutor in legal and technical processes.
    - **Philosophical Outlook:** While social media previously fragmented assumptions into echo chambers, standardized AI responses may create new background shared assumptions among users engaging with similar models. The goal is to transition human function away from labor-based professions toward post-industrial survival, potentially improving demographics by reducing the need for professional specialization.

9. Actionable Takeaways

10. **Follow Expert Analysis:** Sign up for the Amvest Terradin Critical Minerals Fund newsletter at `amvestterraden.com` for deep dives into critical mineral markets, and listen to Thomas Shelley's podcast *Tyranny Today* on Spotify/Apple Podcasts for broader geopolitical and philosophical musings.
11. **Monitor Policy Shifts:** Watch closely for changes in DFARS rules and supply chain regulations coming in January 2025, particularly regarding US procurement from entities of concern (China).
12. **Strategic Planning:** For investors and strategists, recognize that CapEx cycles are being driven by security imperatives rather than interest rates. Prioritize sectors involved in reindustrialization (refining, defense manufacturing, energy capacity) over purely financial efficiency.
13. **Philosophical Context:** When analyzing AI adoption, consider its role in bridging system and life-world communication rather than just as a productivity tool; acknowledge that shared background assumptions may emerge from standardized AI interactions despite the lack of human intentionality in those outputs.

Full Transcript

Welcome to the MacroHive Conversations podcast. I'm Bilal Hafiz, Head of Strategy at MacroHive. Before we dive in, a quick update. Following MacroHive's acquisition by the BGC Group, BGC has launched a new regulated business called MacroHive Markets. For the first time, onboarded clients can also access execution services via MacroHive Markets. This new workflow will allow macro institutional investors to benefit from strategy and quant insights, alongside access to global liquidity. Please contact Andrew Simon or Gail Mozibrian

Bloomberg or click the link in the description below for more details on McRide markets and its service offering. the Amvest Terradin Critical Minerals Fund. He was a former director at the World Economic Forum and vice president of business development at Anglo Gold Ashanti, then the world's largest gold producer. Thomas has spent over 15 years managing commodity and mining equity portfolios for hedge funds and family offices across the US, Europe and Asia. He is fluent in seven languages and a regular contributor to Oxford Analytica. He combines deep industry knowledge with a strategic understanding of how resources, markets and power intersect in today's shifting global landscape. He's also the host of the podcast, Tyranny, today. Now, on to our podcast conversation. Greetings and welcome, Thomas. It's fantastic to have you on the podcast show. Hi, Bilal. Great to be back with you again. Now, one of the things I like about you, Thomas, and I followed you over the years and enjoyed your recent book as well, is that you've looked at commodities and markets for many, many years, and you've got kind of on the ground experience across multiple continents as well.

This year, and in fact, since COVID pretty much, we've kind of almost returned back to the physical world, the real world, rather than just being in this kind of detached financial world. And of course, the Iran Wars kind of really brought things to the fore. So maybe we can start there in terms of how are you seeing the current kind of low level war that's still going on there? There's the hooties of attacking Saudi and the pipeline there. Boats seem to be going through the Strait of Hummus, but we have a refining

issue. There's all these kind of pieces of the puzzle we're trying to put together. But from your perspective, how do you see the energy markets right now? Yes, so before we get to the war on energy market, let me be like, let's zoom out because you pointed out to something really important, COVID, which opened our decade, our unfortunate decade when we have to get used to this new world. But what was the old world?

the old world for about 35 years or so. We used to call it globalization based on certain tenets that were imposed after the Second World War with U.S. hegemony, entrenched after 1991, obviously. There were two parts of this. It was of course a question of security. that reduces the cost of transportation, communication, and so on, of course, makes global trade so much more profitable. And of course, the arrangement around that trade and the free flow, that world has become costly, or the maintenance of the status quo has become costly for two reasons. The first reason is the relative loss of US power. And that is reflected in the thinking of some of the strategies here in the U.S. from West Mitchell to Bridge Colby, who's a senior strategist at Pentagon, viewing the U.S. role is no longer really viable as this global hegemon, but focusing only in areas which are absolutely vital for national economic security. And it has also a geographic corollary. The second reason why we observe that disruption is that technology has advanced the role of revisionist powers. So those that want to undermine that status quo for a variety of reasons, some reasons are maybe purely commercial, but more often than not are geo-strategic and geopolitical. Now the problem with this is that maintenance of status quo is extremely costly.

These are your legacy hardware, blue ocean navies. This means keeping the flow, commercial flow open. It means maintaining a lattice of alliances, naval points, bases, and so on. It is costly. I know that because I pay taxes for that. It is much less expensive at current level of technologies to disrupt those flows. Whether it's at Baben Maldab, whether it's in Hormuz, whether it's in First Island chain, or the Black Sea or the Baltic Sea, it's exactly the same story. What are these technologies? We'll know them all. These are ROVs, so remotely operated vehicles. These are AUVs, autonomous underwater vehicles. These are surface vessels that are also autonomous.

Mines, various agents for anti-submarine warfare. And of course, we've seen activity in the gray zone area, cutting off cables.

both communication and pipelines underwater. It's actually cheap. It's partly deniable. And it makes the maintenance of a status quo so much more expensive. And we've seen I'm not very successful really in some parts of the world. So that's sort of like a global overview of that. And just as we're kind of going to kind of broader context, I mean, how does the U.S. case right now draw a parallel with say the British Empire which was obviously UK was a tiny little island but had dominance in the seas and naval dominance. I mean is there parallels with that and how did the UK lose its dominance? Well the UK was first and foremost undermined by Germany that used exactly the same.

tactics, the asymmetric tactics of disrupting. Kaiser's Germany didn't have any chance to actually deploy navies in a way that would replace the UK dominance at the time, late 19th and early 20th century, but they had U-Bot under Wasserbot, so, you know, submarines that could raid. convoys and maintaining convoys or first world war, especially second world war was a very expensive proposition and the battle of the Atlantic in the second world war was the longest battle of that conflict. So it's absolutely critical, sometimes not understandable for some of your European listeners, but Greenland is critical for that part of the world, Greenland, Iceland and UK, the so-called GI UK gap. Eventually, of course, Britain yielded to the United States that assumed that role. Were the disruptors? Yes, Soviet Navy was a disruptor. And precisely for the same reason, it's based mostly in Murmansk. And there are four fleets for navies to this day, Russian navies. The one in Murmansk is the one that has access to the Atlantic. But the entire theory, and it's good you're going back to that time, entire theory was developed by two strategists. One was Alfred Tayar Mahan, who in 1890 proposed this idea of sort of geopolitical idea of command of the sea, dominant battle, and sort of the control of the critical lines of communication. About 21 years later, Sir Julian Corbat proposed sort of a competing theory that you don't actually need to control the whole thing. It's enough if you actually have impact on what the main lines actually mean, because you cannot really control the ocean. We have terrestrial animals, right? So it's very different to operate on a territory, say Russia is

taking part of another country and then decides what to do with this, populate, depopulate, and so on. Nobody lives on the ocean. And therefore it's very fluid according to Corbett. And at relatively low cost, using those asymmetric techniques, it is possible to undermine the control of the great powers that is called power. So in today's world, when you have so many revisionist forces, even down to the Houthis, which we tend to underestimate, it is possible to sort of trigger a significant disruption of the flow. And I would think that precisely for those two reasons right now, one is the the relative loss of US importance has been going on for quite a while. And now we have this absolute flourishing of new technologies also.

And you mentioned the US of dominance receding. Is that a Trump phenomena or was that happening before that? I mean, a lot of people like to stop in everything to the Trump administration. Yeah, I think it's a very simplistic thing because Trump is a symptom of something that just simply didn't function very well. And that goes back to our loss of industrial productivity over a long period of time. That's closer to my heart and problems of critical minerals and that value chain in general, but also downstream from that. The industrialization of the West was a great idea for as long as 1950s Japanese system of just in time really worked for that unique security. When you don't have security, suddenly you realize that, hey, you know what? We cannot maintain this system because the main producer can use these chokeholds as China did last year in critical minerals. And it's potentially threatening Europe right now during the negotiations to rebalance the lopsided trade relationship between Europe and China. And indeed, there's a conflict between China and Japan. So it goes in many different directions. It is absolutely critical for the West to rebuild that.

capacity. And the reason being is that the strategic doctrines are based on high-tech, short-term war of manoeuvre. So if, for example, something happens to Estonia, then NATO immediately react, crush Russia, and we move on. I think the Ukrainian war and increasingly around war is telling us that war of manoeuvre today and the current Contest between the shield and the sword right and it's a little different online It's a little different and when you fight the sea battle But it's easier and cheaper to send missiles than to find interceptors to stop them Therefore the sword has the upper

hand and therefore the industrial capacity to build the sword is what really matters. Russia is proving this with some support, of course, from China. We hear that the Houthis have support, too, from far away. Wall Street Journal reported on this on two occasions. So that industrial power really matters because war of attrition, a long-term war of attrition, is the name of a game today. And this is what Russia is banking on, and this is what apparently ram is gearing up for.

quite surprisingly. So we can say, OK, it's an asymmetric war of attrition, just enough to block the straight of hammers. This is a low cost. The question to which I do not have an answer is this cost low enough to overcome the problem of 80% or 100% inflation that ran? This, of course, I don't know. It's a question for their own rulers. And so in terms of the dynamics of the energy markets this year, what we saw earlier this year during the initial phase of the war. The US-Iran war was that there was massive concerns about crude oil. It was kind of a crude oil issue. And then somehow we didn't end up with this massive spike in oil that crushed the world economy. It seems like there was some combination of reserves being used and China importing less. So somehow we kind of muddle through all of that. Now the focus seems to be on refined products, you know, with correct spread so wide. And so there's a question around who has refining capacity. Is that kind of the right way of looking at...

the energy markets right now? Am I missing something or perhaps you could expand on this? No, I think you're right on target in this. If we just roll back to how the market behaved back in March, you may remember that there was rightly so a lot of concern about crude oil and people forgot about refined products or indeed aluminum or sulfur, sulfuric acid and fertilizer and other things, right? And that had its impact later on, sort of knock on impact on many of those. However, initially in March, if you look back, the futures market didn't show anything like 100 per barrel. So there was an assumption that this 1991 sort of Gulf War style big punch will solve the problem relatively soon, that at least this is what the future market was telling us. Of course, six, seven months later, that's no longer the case. And you write about the refining capacity. The problem of refining capacity is that for several years really, you know, our sort of green push towards new energy sources resulted in very little capital going into new refining

capacity. Of course, we have a lot here in the United States in the South that's sort of not necessarily attuned to the type of crude debt that's flowing there. But in the West generally, there was very little.

new capacity, there is quite a lot in China, in southern China, there is quite a lot in Russia as well. Now, of course, unfortunately, today, Russia and China are no longer really a reliable player, reliable players. China, because the knee jerk reaction of the CCP, when any of these sectors that I mentioned is threatened, China's reaction is, well, let's constrain exports. And that happened, of course, to certain fertilizer. It happens to refined products. It happened to sulfuric acid as well. Russia's problems, of course, different because here Ukraine is in a position of the disruptor, right, of this sort of trying to impose a denial for Russia's sort of economic prowess. Whether that has a really impact on, say, US diesel market, most people doubt about it. And also the question is if there is a diesel deal, is the Russian government ready to leave?

poor Russians waiting in line for their products and not having them just because it has to flow across the Atlantic, which is also not necessarily something that will be felt where it should, which is, say, Midwest or somewhere there politically. So I think it's coming back to that issue of reindustrialization and energy problems in Europe and energy dependency and so on. If we have the world that the minister lover of dreams of, multi-polar world, with these disruptions constantly, then of course, each of those longer term relationships are under a question mark. And energy flows are as well. And so rebuilding refining capacity that will be one of the big challenges for those parts of the world where there is certain deficiency and over dependence on global flows.

I'm increasingly expensive. I mean, just think about insurance for shipping and not just in those open lines. I'm concerned about Kazakhstan, which has often one of its roads is through the Caspian. Well, Caspian has also become now a theater of attacks. for example, Iranian assets there. And so that boosts the insurance to ship different products out of landlocked countries such as Kazakhstan, which previously already had problems shipping on products to Russia because of sanctions. So it is major dislocation. And what does it take to take it back? Well, of course, the end of the current

kinetic conflicts would help in the short term. But this is not a reason to then just fall back and think that we're back in 2019. We're not.

And I guess one of the recurring themes this year has been that, given all of these problems, why hasn't say somewhere like Europe, which is an energy importer, why hasn't it been crushed into a recession or India or all of these importing countries? Why hasn't that happened? Well, one thing is that energy prices haven't really skyrocketed as people expected, right? So partly that's because of, let's call it demand destruction, but it's a political decision by China to build up this massive strategic petroleum reserve. We don't know exactly how much it was, but it was all close to twice the level of US before the war started. And so China just withdrew from a lot of these markets that helped the prices. Secondly, of course, we have a very strange sort of pull right now between the rebuilding of the capacity and that's actually advancing on the defense side pretty quickly. You'll find a number of, say, German companies that used to just provide components for the troubled auto markets. And suddenly, many of them are moving towards rebuilding of defense capacity that comes, of course, with a lot of public money on day one. But it's a source of growth for some of those economies in Europe. And, you know, in many other places, productivity is growing, productivity is actually rising with application of AI. And we should not underestimate it. Now, what it means for a labor market and other issues, it's a separate story. But today, definitely, there's a strong push. So between those and the fact that Europe has been kind of you know, growing very slowly for quite a while. I think that push towards rebuilding certain capacities, predominantly defense day one, but also some civilian that includes energy and so on. That's sort of ironic because of, if you remember, after 2010, 2011, emerging from the global financial crisis here and then European fiscal crisis, despite extremely low interest rates, there was very little capex in the West, right? Whatever capex there was, it was in China and capital fled to China. Capital doesn't necessarily flee to China so much anymore to build capacity. And therefore, there's much more investment in rebuilding and reindustrializing the West despite high interest rates. It's actually a very interesting lesson. It means that interest rates are not the driving force of the CapEx cycle, something else's. Yeah. And say on the, you know, whether it's crude refining capacity, I mean, what are the things we need to look at for signs that things are going to ease up? One, as

you said, obviously, reduction in the kinetic aspect of the conflict. What else do we need to look at to kind of get a sense of good or bad on that side?

Well, I tell you, it was really good. I mean, there's a lot of capacity coming in the Western world, so in the Western Hemisphere. If there is a winner of this war, this is gonna be the Western Hemisphere oil production. Brazil, Argentina, Venezuela, Guyana, US, Canada, it's astounding. It's very likely that they will. one CNMR, the dominance of the Persian Gulf producers, as much as we did until this year. And that will have very significant strategic consequences for everybody, because then you have to build capacity for that product specifically. I think Europe will be the natural client there. Many of these are on the Eastern seaboard, so it's a little easier to move it then, say, for Asian clients.

Asians have a problem. Northeast Asia has a problem, of course, and that's because of a very different conflict over the first island chain and access to Persian Gulf. And because of what's been going on in the Panama Canal, I do not think that this increase in capacity for energy production on the eastern seaboard of the Western Hemisphere will easily solve the problems of, say, Japan, Taiwan or South Korea. Okay, and then on the refining side, you know, we still hear about China has a lot for refining capacity. There's concerns that, you know, they are restricting exports. But is China the swing variable on the refining side, or can the Middle East or other regions pick up sort of refining capacity? In normal circumstances, of course, the Middle East could pick up that capacity. The problem is...

China being so trigger happy restraining its exports on anything the moment there is a conflict somewhere else. This is not a reliable supplier anymore. And so it's very difficult. On the other hand, of course, we see the reticence of the United States to impose sanctions on those teapot refiners in southern China. for a variety of other reasons and leverage that China has in other sectors, not least my own sector. So if you think about independence and resilience in global energy markets, putting your axe into the likes of China, Russia, Persian Gulf might not be the ideal solution going forward. So we're back to the Western Hemisphere really. And, you know, China always seems to come up as such an important player in all of this. So when you look at, if you step back and look at sort of China from a geopolitical

perspective, geostrategic perspective in terms of its, you know, tussles with the US, you know, the way it's conducting itself with this trade deals with Europe. I mean, how do you see what game is China? What is strategy here?

Well, first, it's not in China's interest to see Russia lose the war, right? So the longer the war lasts, the better. That reduces Russian power, of course, makes Russia more dependent on Chinese components and inputs. It keeps Europe divided, because this war, let's face it, divides Europe in a significant way. So that's very helpful for China. For example, during those trade negotiations, it also distracts the United States less now than under Biden administration, but still. away from, say, East Asia. So all of this is extremely positive for China to continue supporting Russia, not to escalation, say, threatening nuclear, anything like this, because that will be against China's interest. But at the current level, it's extremely helpful. This is why China, for example, slapped the sanctions against European companies last time where Europe, I think, was 20 seconds.

round of sanctions against Russia that of course in this net there were a couple of Chinese companies that were caught which provide lethal inputs for Russian industry and China immediately on the same day or day after reacted by slapping sanctions on those companies, those tech companies in Europe and indeed some academic institutions that provide Europe with the capacity of deterrence against Russia. especially not only just during the Ukraine war, but also those various gray zone activities and potential also horizontal escalation. Some people fear Russia could instigate. And so those Chinese sanctions really affected not only the iron metal, which is not obvious. target in Germany, but also, say, Wroclaw University in Poland, which works on some optoelectronics that could be potentially used with dual use applications. So very strong, very strong pro-Russian, very immediate.

reaction. That's something that we didn't see two or three years ago at the beginning of this conflict in Europe. And most analysts, and I think I agree with this, attribute this to the fact that China feels bolstered and strengthened by winning the trade war of the United States in 2025 and forcing the U.S. administration to genoflact in front of John Donkai in order

to obtain critical minerals. That issue is not over, of course. We're just kicking the ball down. down the road. And so the way in which China is acting vis-à-vis Europe and indeed Philippines, Japan, Taiwan and others in Eustasia is a good example that that sort of level of self-confidence has grown since last year. And just on, I mean, this is a bit reductive, but is, you know, say China versus the US, if we kind of view this as a battle between these two, is China, is it winning or not? Or was that too, I mean, it's a bit of a silly question at one level, but.

It's just to paraphrase Joanne Lai speaking about French revolutions too early to say, right? Here's the way I view it. So those negotiations, and I think we spoke on your podcast before about it, negotiations in Busan last year in November, granted the West, well, it's just not the US, but granted the West a one-year reprieve in terms of the tightest restrictions on critical minerals, especially rare earths. Now, there is a system in China, which is extremely restrictive of control of these flows. It's kind of like a valve that they can open and close and open and close and open and close. And coming into the most recent meeting between the two leaders in Washington a couple of weeks ago, the Chinese side proposed to extend that reprieve for another two years. And this time really with relation with the US because conflicts with Japan and Europe just have a different dynamic right now. And this was rejected by the US side.

And I find it a fascinating moment in this struggle. Why is that? Because you can view that value of that leverage for China as sort of a long option. And in option trading, the value of the option depends, of course, on how far you are from the strike price, how far you are from exploration and the volatility. So we don't know about the strike price, but we know about the timeline and the volatility. The volatility depends on China opening and closing the tap. And of course, if China proposed two-year extension to the end of Trump's term, that of course increases the value of the option. U.S. decided, no, let's just keep it till January. In January, what happens in January is that we have a tightening of so-called DFARS rule, which is Defense Federal Acquisition Regulatory Supplement.

So that tells various procurement departments and various companies that eventually serve the defense department, that they're not allowed to source

any materials from the foreign entities of concern that includes Iran, China, North Korea, and Russia. That means tightening of this, because that always existed, but there are some carve outs. We cannot get graphite, so please allow us to get some from China. So in a way, when US bans itself and shortens the duration of the option, the value of this leverage, false. And that's probably come as a surprise for the Chinese side. We'll see whether the US can actually live up to its own self-imposed constraint in January. And that really affects just the defense department and the defense side of the demand, not the civilian one. And so I tend to compare the two different games in game theory. The idea is that in 2025 when China was decided to use this particular escalatory model, it was like a game of chicken, you know, like in the old movie with James Dean, right? The two cars that just go at each other, right? And, you know, you're in bad shape if you're the first one to yield and swerve because, you know, women gonna laugh at you and say you're a chicken. But it's worse if both actually crash, right? And that contest in 25, China decided that the cost of crashing for it was actually much lower. The pain threshold was much more comfortable than for the United States. So say the US pain curve was much higher. And that's because of the leverage that represented in downstream industry, basically kneecapping entirely the parts of the US industrial base. And so US had no choice but to yield. And that changes the game from the game of chicken to game of bully. And China was capable of escalating this again between April of last year and October of last year, and then maintaining this option for longer term. This change with DFARS potentially blocking ourselves by our own decision from buying things from China, separates this into two different games. So on the defense side, we're back into Game of Chicken, and we'll see who yields in January, or whether UIS has maintained or rebuilt some capacity to just feed that particular sector. We'll see. On the civilian side, we're still in a bully game, and China is definitely willing to use the Valve and increase the volatility of that option as it pleases.

So do your question, who's winning? It's hard to say. We're in the middle of that. Can you just on that, you know, this self-imposed ban? I mean, you said you alluded to, I don't know the details of what critical minerals or rare earth, you know, that the US can't do, but... Where can they get graphite you mentioned or like if not from China? I mean can the US source all of these things somewhere else in the world? It's just that previously it was

expensive for them to do so or what's kind of the deal there? I mean they surely they need some of the stuff from China. Right, absolutely. So graphite is a particularly fraught.

a problem, where for natural flight graphite sphronization is almost still entirely controlled by China. Now, I heard very interesting stories, for example, how the South Korean defense industry manages to go around that, of course. Well, they created very fruitful relations with some Russian counterparties and some counterparties in the Middle East. who have friendly relations with China and obtain this material and are happy to earn a little more. You know, hard currency won, in this case, to smuggle things out and allow South Korean defense industry, which is a large part of its economy, to obtain the components that China wouldn't sell directly to South Korea. So, you know, there are ways to obtain things. Now, I'm not sure whether that affects graphite. On the rare earths, there's been some progress, not necessarily yet in the United States, but I just saw pictures of the new plant in southern France.

I run by Carrister. It's a solvent extraction plant. It's amazing what they managed to put together in just three years. So that level of dependency that we saw when China started this probably in 2020 with the expert control law, that probably is peaking right now. And it's always a risk when you kind of use and abuse the same leverage over and over again. People talk about US using the dollar. this way in sanctions, building by default those areas of bilateral relations, trade relations that are independent of the dollar. This is something similar. You can only use this leverage up to a certain point. Now, this is not overnight. That particular plant in France is not going to solve all the problems of the West, but it will help and there are many others coming.

you know, in four years from now when we have this conversation, it's going to be a very different world. And, you know, earlier we talked about Russia and we talked about it, you know, in the context of China as well. But like more specifically on Russia and the Ukraine war, you know, this has been going on for so many years now. There's some obviously the US is kind of reducing its support for Ukraine within Europe. There's a division occurring, especially the most right-wing parties are questioning whether

they should support Ukraine. At the same time, there's some concerns about the Baltics as well, whether Russia will kind of target those countries. So what's your thoughts on how we should look at the Russia-Ukraine War? This is more than Russia-Ukraine War because the objective for Russia was spelled out by its Ministry of Foreign Affairs in December 21.

And that is what Thomas Shelley used to call compelence, forcing the other side to do something. Now, what is something? To move NATO forces away to the pre-1997 line in a way to allow Russia to reestablish its sphere of influence. Russia harks back to this Tsarist era when not only the Baltic countries but Finland and Poland were also part of the Russian Empire. And so they would love to reestablish that. Ukraine is just... you know, one part of that puzzle. Williams, I mean, of course, tragic was going on there on a nightly basis. But the gray zone tactics has now spilled over widely beyond Ukraine, right? Romania, Bulgaria, Poland, the Baltics, Germany, Scandinavia, and so on. The two characteristics of this one is repetition, and sometimes escalator repetition. And the other one is what I call implausible deniability. Sounds like Semaronic, but stay with me for a moment. It's plausible.

and implausible at the same time. Russia can deny that it did any of that, jamming GPS, cutting the cables, arson attacks or homicide and so on, different types of threats and can deny it in the courtroom. But it makes it in a way so visible and so obvious that everybody in the public sphere realizes that it's Russia's actions. So there are two sides to this. And it does a couple of things. First, it maps the thresholds or the other side would eventually push back, right? Are we going to push back against GPS jamming in what way? But okay, so there's a airspace violation. How do we react to this, right? The second part of this is of course normalization. If something happens over and over and over again, like China crossing the midline of the Taiwan Strait, right?

It's no longer news, right? It's something that happens and people are numb to the news that sort of loses its shock value. And the third one is, of course, what you mentioned, divisions, both geographic divisions, because there is asymmetry of fear between different parts of the European Union or Europe in general. And of course, internally as well, where the extreme left and

extreme right is very well incentivized to believe that this is not a threat to their own polity, say in Germany and France and some other places. And that's because of what I call the paradox of weak power. During the First Cold War, the USSR was so powerful that people in Paris could actually be concerned. And also, there was an ideological link to certain parties there directly in Italy, especially in France as well, that Soviets had a lot of influence over the political scene there directly. This is no longer the case. Russia is not sufficiently strong. And therefore, it's really threatened, say, Estonia, Latvia, Lithuania.

Moldova and then maybe Poland, Finland and so on, but not Portugal, not Netherlands and so on. And so there is a question of proportionality. How do we actually respond to these actions? What does it mean if little green men appear in Narva in Estonia? How to react to that? Is there proper doctrine to deal with this? Let's not forget that an alliance means something very different during the times of peace. and when the shooting starts. One is a political diplomatic decision, and it should actually act as deterrence against an adversary. But the moment the shooting starts, it's very different. We know it from 1930s. The alliances, the real test is at that time in Madrid, are we going to send our sons to die on the other end of the continent? And of course, this is why the gray zone tactic is so helpful for that repetition and slow escalation because Russia can constantly test how far it can go.

before a new Spanish prime minister has to decide that, in fact, this is something that's at stake even for Spain. We're far, far, far away from that. OK. I mean, that's quite a worrisome development there. And is there... Sorry to say that. It's worrisome because at the same time Russia is capable of producing 3,000 jet drones a month and over 100 ballistic missiles. And right now Ukraine is on the receiving end of that. Europe has nowhere near that capacity. And so that goes back to my first answer to your first question. There is no other choice than to spend half a trillion or a trillion dollar a year to rebuild the industry. And the problem in Europe specifically is because Unfortunately, the budgets, with the exception maybe of Germany, are extremely strained. We know it from the French curve, the Yield curve, UK is not in a better space, and so on.

you know, where to take the the means to actually increase the security because without security, there is no economic growth possible. And so, you know, the different views of, you know, different, you know, Masayoshi, someone believes there's going to be five trillion dollars a year in terms of, you know, AI buildup and so on. That's maybe on the extreme side. But we're seeing here, definitely in the US with the new hardware going into data centers and hopefully replicating this into a physical fabrication model. and robotics where US is quite behind China at the moment. That is absolutely necessary. We have to re-consolidate the entire manufacturing process rather than being at the forefront of R&D and then seeing our capital exporting all the capacity to scale it up in China. It's no longer really plausible because China is not only putting sanctions on European companies but also supporting Russian effort. Speaking of data sensors and AI, what are your broader thoughts on you know, data centers and AI. I also know you've got some really interesting philosophical views as well. So we can probably get into that as well. But broadly, you seem kind of, you know, advocating for the data center build out and this kind of whole infrastructure side of the story. But more broadly AI, what's your thoughts? I mean, in the markets, quite. There are three times two, probably six ways to look at it, right? Optimistic, middle and pessimistic and anthropocentric and non-anthropocentric.

of the things that I haven't spoken yet, but I'm happy to share with you and your listeners. I recommend you read Stanislav Lem, a Lviv-born science fiction writer in the 60s, who wrote a couple of wonderful stories, Golem XIV, The Invincible Solaris, which I think was filmed on a couple of occasions, in which the thing, the thing which is not us, with potentially consciousness and potentially with intentionality, is completely indifferent to our fate. It's like weather. It doesn't care. So it's neither friendly nor hostile. It's maybe the best attitude we can have to the universe in general. But here it is. Something is coming up. And essentially, people still think it's a tool. Malcolm McLuhan, remember, in medium is the message. He believed that the real thing is the medium. And here's the medium that's increasing autonomous with the genetic risks, with potential gold drift, that's the intrinsic risks and potential outcome problems, which is the indirect risks. And everybody is looking at the risks. I only know that my book that you read, I could translate into Japanese in just eight hours, just

simply spending two hours of my time overviewing all the different metaphors that I had to fine tune, right? Because they were imperfect. That's an amazing time creator. That's a time producer.

But the context, the broader context I used in my recent talk was about applying communicative action, Eurigine Habermas's concept. He borrowed things from phenomenology, from social interactionists, from Frankfurt School and so on. He viewed our world as sort of that, you know, on the one hand, there's the life world, so Lebenswert in German, which is this intuitive, pre-scientific, everyday core of shared assumptions, right? like you and I have certain certain assumptions, we can communicate this way and we communicate with language. But there's this other world, which he back when he wrote it in 1981, is the world of the system, the macro system. So it's both the powers of bureaucracy and the economics, mostly money, right? So it's the linguistic fight, right? It's not language. It's in other ways it impacts on how you're functioning. He was concerned that the system was colonizing Laban's belt or life world. And then he tried to connect the bridge and found the law maybe legal system is something that bridges the tool. And the question is like, so where is AI? Is it here? Is it here? Is it a bridge? And my answer to this, it's actually everywhere. It's everywhere. It's part of a system because it allows you to say formulate complaint or understand the diagnosis that it can actually create the legal material and so on. But it's also interlocutor.

translate my book, I discuss with this. So maybe it's better to do this. What do you think? And so on. And so it's sort of a mutual learning process. And that's unprecedented. We never had something that functioned in all of these polls at the same time. And because it's a language system, one thing that you can ask doesn't make sense. It's like, who are you to say it, right? What's at stake for you? We don't know. There's this lack of intentionality. And that's why Stanislav Lam's science fiction is actually extremely useful. Because maybe that thing just simply doesn't have intentionality. That's anthropocentric for us. I'm actually falling into this more optimistic side of the various philosophical... Why so? Why? Why? Why so? Because there was a man with whom Habermas would definitely disagree. His name was Arnold Gehlen, G-E-H-L-E-N. He wrote in German about Entlastung. Entlastung. So unburdening.

It's taking things away from us that we're burdensome. Accounting is burdensome. Modeling certain things is burdensome. And I lost my eyesight on Excel building models and financial models. And many people do jobs that they do not necessarily enjoy. So it is quite possible that our function in society will be something different than a profession. because so many professions will be taken away from us now. We have to solve the problem of income and, you know, survivability and so on. But we had different models in history, right? I mean, there was a time where, you know, ancient Greeks had those things that were doing all the jobs that were called slaves. And therefore, Greeks could, you know, on Socrates and Plato and Aristotle could just cogitate and create the foundations of the Western society thinking. The Romans had enough time, for the same reason, more or less, to create public sphere, the Senate, the institutions, and so on, that led eventually to Roman law, EUCVDs, and then confusion with kind of law-created civil law. I mean, didn't we just say, in kind of the Habermas of, you know, world, if the life world, the Laban's Belt is being invaded by AI, then don't we lose something then as humans? Maybe we gain something. So let's again, let's look at it. And we've been invaded by social media for so long now, right? What it is, 25 years or something. Yeah. Dismembering ourselves into those echo chambers.

That's the foreground of Lebensfeld. But if sufficient number of people goes into the same five or six models right now that dominate, right? And ask about the specific medication or ask about a specific political candidate or something. And the answers are trained and come back with similar answers all the time. So you know what it is? It's no longer the foreground. We actually gain shared assumptions, background assumptions, yet again. Some people say, oh my God, I would love the days of New York Times generating the political reaction in the United States against Nixon rather than having these multiplicity of different. conspiracy theories and such like on TikTok and so on. So maybe we're seeing actually an emergence of background shirt assumption thanks to the fact how that functions in response to that dialogue. What I am concerned about is that some people do not view it as a process, but as a pure output. And that's a concern for young people who have never had the chance that you and I had building our world in 3D human interactions through a proper physical experience. They jump directly from childhood into adulthood with the presence of this

and may expect this to just provide output rather than engage in a process where I teach this thing something and that teaches something back to me.

But in the broader, I come back to what I was saying before, is that the society could be completely appended. The role of women could come back to what it was before industrialization. Who knows? If we don't define ourselves by professional function, but by something else, within the family, within some other social context, that's not necessarily all negative. It's not necessarily all negative, because if you look at our demographic growth right now, if we don't do something about it, we will be wiped out by our own decisions of not having children. So maybe AI will help us actually move away from this and change the way we function, not overnight, but in the longer term, it's quite plausible. Well, I think on that note, it's a great way to end the conversation. I mean, we could go on more. Just a final thing. What's the best way, Thomas, for people to follow your thinking, your work? How can they reach out to you?

So there are two facets to this. One is for people who are really interested in what I do professionally, which is critical minerals fund. And there is a weekly newsletter that sort of delves into this and more detail. It's a lot more technical, but maybe something of interest. That's amvestterraden.com. You can sign up to this. And for my broader musings, it's always Terranee Today, my usually weekly podcast, Terranee Today on Spotify, Apple, and so on. I definitely recommend people to sign up to the first and listen to the second. And on the second, on the podcast, you can also often hear Thomas singing as well, which is always fun too. So with that, Thomas, thanks again for this really entertaining conversation and keep up the good work. I mean, you really have an original take on what's going on in the world and a very valuable one too. Likewise. I love your podcast. Thanks for doing this.

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